# HIMS Documentation

Health Insurance Management System — Project Technical Overview

## 1. Project Overview

The **Health Insurance Management System (HIMS)** is a modular enterprise solution designed to manage the entire insurance lifecycle. Built with a focus on scalability and user experience, it serves as a central hub for Customers, Agents, Admins, and Claims Officers.

From the initial premium calculation (Quote generation) to the final claim disbursement, HIMS ensures data integrity, transparent tracking, and secure document handling.

## 2. Modern Tech Stack

### Backend (Core Engine)

- ASP.NET Core 8.0 / Web API

### Frontend (User Interface)

- Angular 17+ (Modular Structure)

- Entity Framework Core
- SQL Server (Database Management)
- JWT Authentication & Role Control

- Reactive Forms & RxJS Services
- Custom CSS Design Architecture
- Dynamic Dashboard Rendering

# 3. Entity Variable Reference (Entities Folder)

Detailed mapping of variables within the `HealthInsurance.Domain.Entities` namespace.

*Note: All entities share a common* ***`BaseEntity`*** *providing Id, CreatedAt, UpdatedAt, and IsActive fields.*

## 3.1 User Entity

| Variable | Type | Description |
|---|---|---|
| FullName | string | The legal name of the user account. |
| Email | string | Unique email used as the primary identifier. |
| PasswordHash | string | Encrypted string for authentication security. |
| Role | UserRole | Enum: Admin, Agent, Customer, ClaimsOfficer. |

## 3.2 Policy Entity

| Variable | Type | Description |
|---|---|---|
| PolicyNumber | string | Unique public identifier for the contract. |
| PlanName | string | Title of the insurance product. |
| MonthlyPremium | decimal | The calculated cost payable monthly. |
| CoverageAmount | decimal | The maximum liability of the insurer. |
| ExpiryDate | DateTime | The date the policy coverage expires. |
| Status | string | Contract status: Active, Lapsed, Pending. |
| AgentId | int? | Reference to the agent managing the policy. |
| IsPlanTemplate | bool | Distinguishes templates from active policies. |

### 3.3 PremiumQuote Entity

| Variable | Type | Description |
|---|---|---|
| ProspectAge | int | Age of the applicant used for premium calc. |
| SelectedPlanName | string | Plan selected by the prospect. |

| Variable | Type | Description |
|---|---|---|
| CalculatedMonthlyPremium | decimal | Final calculated cost. |
| QuoteReference | string | Unique 8-char reference for lookup. |
| IsConvertedToPolicy | bool | Indicates if a purchase was finalized. |

## 3.4 Claim Entity

| Variable | Type | Description |
|---|---|---|
| ClaimAmount | decimal | Financial compensation requested. |
| Reason | string | Detailed text description of the event. |
| Status | string | Workflow state: Pending, Approved, Rejected. |
| PolicyId | int | FK identifying the source policy. |

## 3.5 DocumentVault Entity

| Variable | Type | Description |
|---|---|---|
| DocumentType | string | Type classification (Aadhar, Hospital Bill). |
| FilePath | string | Physical storage path or URI. |
| Status | string | Verification status by the Claims Officer. |
| RelatedEntityType | string | Generic link (Policy/Claim context). |

## 3.6 Payment Entity

| Variable | Type | Description |
|---|---|---|
| UserId | int | Link to the user who made the payment. |
| QuoteId | int | Link to the source quote being paid. |
| Amount | decimal | The total currency amount processed. |
| PaymentMethod | string | Method used (e.g., CreditCard). |
| TransactionReference | string | External tracking ID for the transaction. |
| Status | string | Payment status: Completed, Failed. |

## 3.7 CommissionLog Entity

| Variable | Type | Description |
|---|---|---|
| AgentId | int | The agent recipient for the commission. |
| PremiumAmount | decimal | Monthly premium triggering the payout. |
| CommissionRate | decimal | Percentage multiplier (e.g., 0.10). |
| EarnedAmount | decimal | The final calculated commission (Premium * Rate). |
| Status | string | Payout cycle state: Pending, Disbursed. |

## 3.8 ActionLog Entity

| Variable | Type | Description |
|---|---|---|
| EntityName | string | The target entity (e.g., "Claim" or "Policy"). |
| EntityRecordId | int | The specific ID of the target record. |
| ActionType | string | The operation performed (e.g., "StatusUpdate"). |

| Variable | Type | Description |
|---|---|---|
| OldValue / NewValue | string | Audit trail for state changes. |
| PerformedByUserId | int | The Staff member who executed the action. |

# 4. Core Algorithms

**Premium Pricing Calculation:**
The system utilizes a risk-adjusted model to determine monthly premiums:

```
FinalPremium = (Base_Rate + (Age_Load)) * Tier_Multiplier
```

**Tier Multipliers:**

- **Silver:** 1.0 (Baseline)
- **Gold:** 1.2 (+20%)
- **Platinum:** 1.5 (+50%)

# 5. Workflow Overview

HIMS operates on a linear workflow with branching permissions:

- **Onboarding:** Guest/Customer generates a `Quote`.

- **Conversion:** Quote is paid and becomes a `Policy`.
- **Verification:** `Agent` verifies the policy details.
- **Servicing:** `Customer` raises a `Claim` and uploads documents.
- **Auditing:** `Claims Officer` validates documents and updates status.

---

© 2026 Health Insurance Management System. All Rights Reserved.

This document was automatically generated for project documentation purposes.